Masterthesis Proposal:

# Comparison and Development of Tree Height and Biomass Allometries based on TLS Data of Forests in Ghana

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Potsdam, 17 June 2026

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# Introduction

- **Tree height** and above-ground **biomass** (AGB) are **key variables** in forest studies
- **Direct measurements** are often **inaccurate** or too **time-consuming** for large-scale applications
- Therefore, allometries are widely used
- **Allometries** describe **relationships** between easily measured variables (e.g. **DBH**) and difficult-to-measure variables such as **height** or **AGB**
- This allows comparatively **simple estimation** of the more complicated parameters
- Relevance for **Ghana**
  - Key ecological, economic and social role of forest
  - Strong **decline in forest cover** since the 1980s
  - Need for reliable biomass and carbon estimations

# Theoretical Framework

- Large variety of allometric equations
  → different predictors, datasets and applications
- **Global models** (e.g. Chave et al., 2005)
  → widely usable but lower local accuracy
- **Local/regional models**
  → better representation of local conditions
- Few allometries available for African forests
- Frequent use of global/regional models in Africa
  → higher uncertainty
- **No** nationwide **allometric model for Ghana**

- Allometries are **based** on **measured data**, from which these relationships are determined
- Datasets ranging from **~50** to several **thousand trees**
- Most studies use **destructive sampling**
  - Trees harvested, dried and weighed/measured
  - Very accurate, but high effort, costs and environmental impact
- Increasing use of **LiDAR-based** approaches (TLS/UAV)
  - Parameters extracted from the point clouds
  - Lower effort and no destructive impact while maintaining high accuracy
  - Enables large-scale studies with fewer resources

→ Therefore, I use terrestrial LiDAR data from Ghana

# Objectives

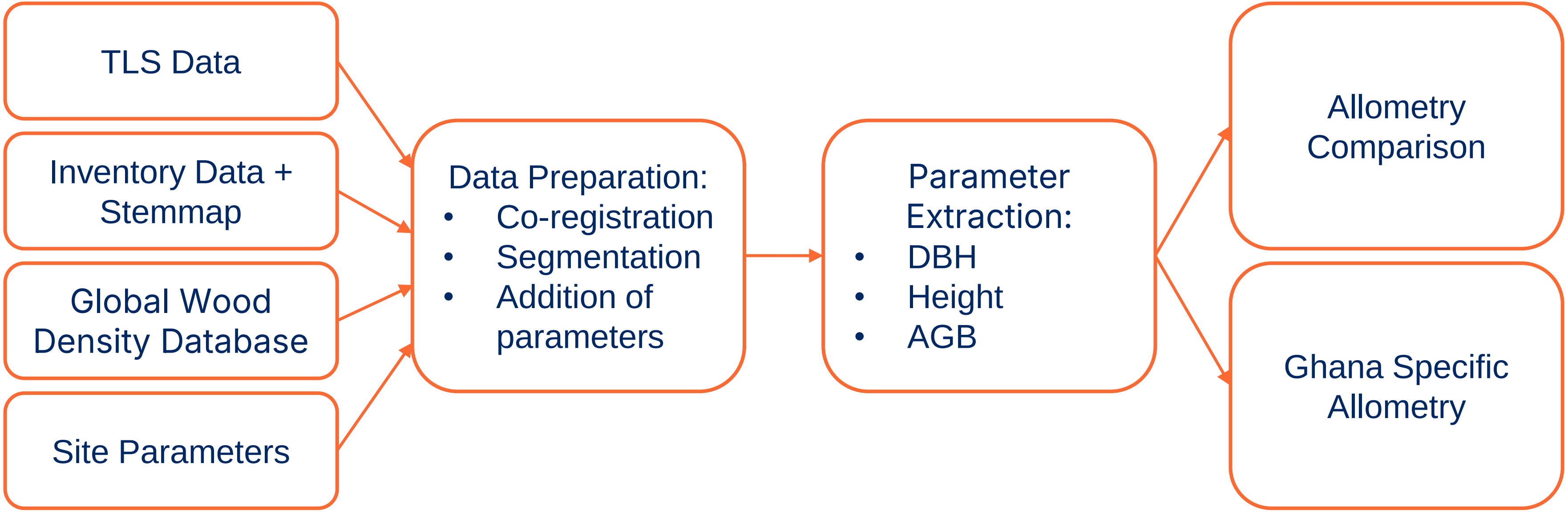
**Comparison** of **selected allometric models** using TLS-derived structural parameters as reference data

Analysis of **structural** and **environmental influences** on DBH–height and DBH–AGB **relationships**

Development of specific **height and biomass allometries** for **Ghana** based on TLS data

# Workflow

To address the defined objectives, the following workflow has been applied and will be continued in the next months:

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# Data
## Study Sites

6 Plots:
- Ankasa 1 (ANK1)
- Bobiri 1 (BOB1)
- Bobiri 2 (BOB2)
- Kogaye 2 (KOG2)
- Kogaye 4 (KOG4)
- Kogaye 5 (KOG5)

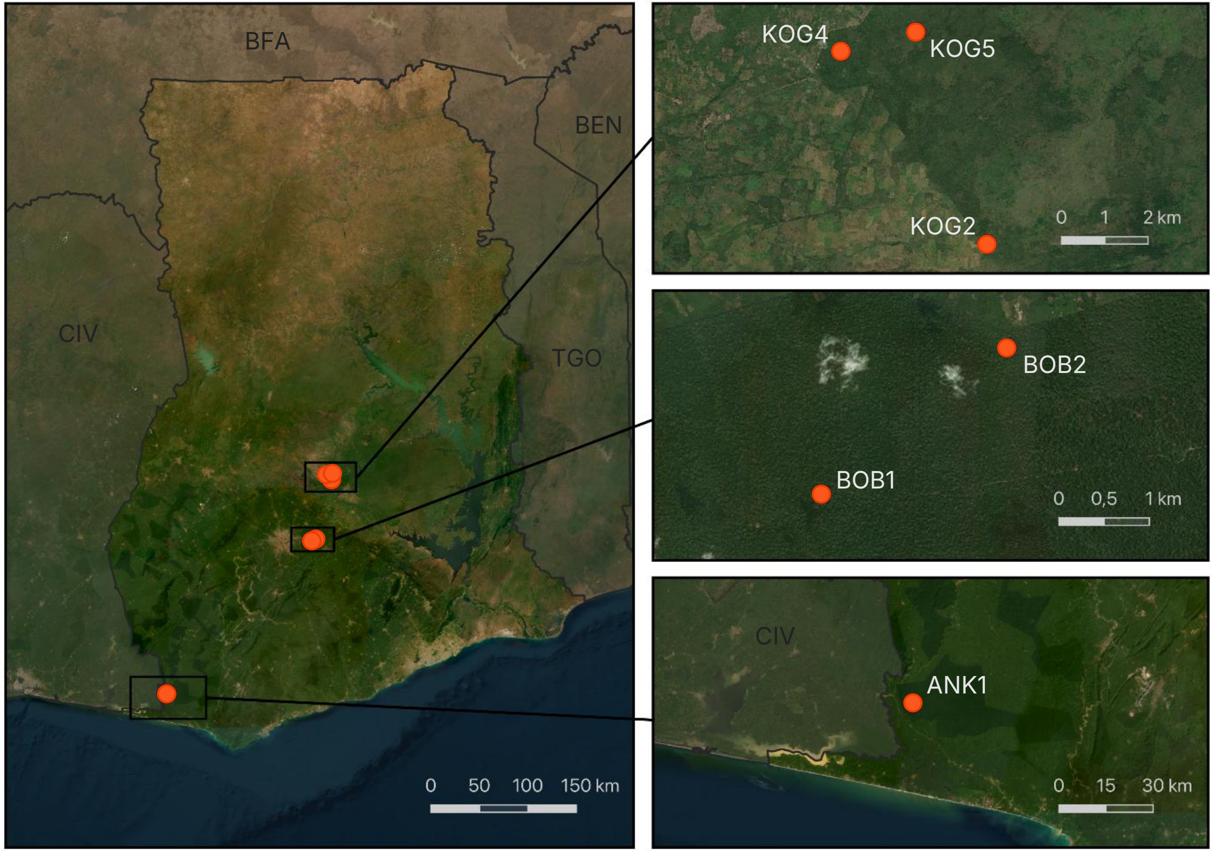


Fig. 1: Location of study plots in Ghana

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- Vegetation in Ghana strongly shaped by a **north–south rainfall gradient**
- Humid tropical rainforests in the south transitioning towards drier savanna landscapes in the north
- Study Plots cover a broad range of major vegetation zones:
  - ANK1: **wet evergreen forest**
  - BOB1 & BOB2: **moist semi-deciduous forest**
  - KOG2: **dry semi-evergreen forest**
  - KOG4: **forest–savanna transition zone**
  - KOG5: **wooded savanna**
- No TLS data available for northern savanna regions → limitation of the study

→ **Good representation of Ghana’s forest ecosystems**

# Data - Study Sides

Fig.2: Sample photos of different plots, left to right: KOG5, KOG4, BOB1 & ANK1
(Photos: J. Wilk & N. Laudenbach)

# Acquisition

Data acquisition using **terrestrial LiDAR Scanning** (TLS) Scanners used:

- RIEGL VZ-400i
- RIEGL VZ-600i

Multiple scans performed within each plot to create one point clouds

Scan spacing depending on vegetation density:

- **20 m grid** → Kogyae plots
- **10 m grid** → Bobiri & Ankasa plots

**Two scans per position** with different scan angles, for less occlusion

Fig.4: Scan with the RIEGL VZ-600i (Foto: J. Wilk)

# Data Preparation

- **Registration** of individual scans into plot-based point clouds using **RiSCAN**
- **Statistical outlier removal** to reduce errors and noise
- **Tree selection**:
  - targeted total: 80-140 trees
  - DBH > 10 cm
  - high point cloud quality
  - balanced representation of: tree sizes, species & vegetation zones
- **Tree segmentation**:
  - coarse segmentation with RayCloudTools
  - manual refinement in CloudCompare

**Parameter Addition:**

- Additional parameters included to **improve explanatory power** of the developed allometries
  - **Inventory data** (FORIG): species, tree condition, additional tree attributes
  - **Site parameters** (Thomson et al., 2018): precipitation, temperature, soil conditions, drought stress, elevation
  - **Global Wood Density Database** (Zanne et al., 2009): species-specific wood density
  - **Competition and neighborhood effects**: local tree density, BAL index, Hegyi index

# Extraction of Structural Metrics

Key structural parameters: **DBH, tree height & AGB**
Parameters **extracted** from segmented TLS point clouds and used as reference data

**DBH:**

- Extracted with the **ITSMe package** (from Louise Terryn)
- Tested and established functions for robust DBH estimation
- Works also for complex and irregular trunk shapes

**Height:**

- **Difference** between highest and lowest point

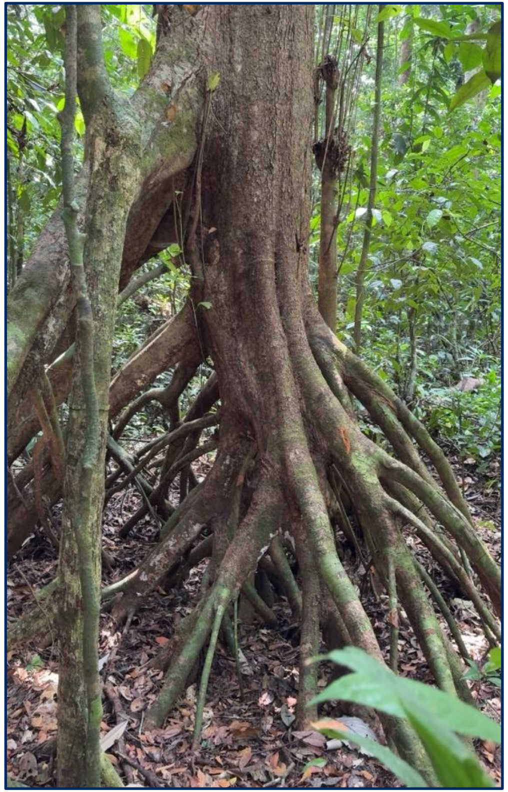

Fig.4: Problematic Tree
(Foto: N. Laudenbach)

**AGB:**

- **Quantitive Structure Models** (QSM) generation using RayCloudTools
- Tree structure approximated by cylinders
- Leaves are not included
- Biomass derived from **volume × density of tree species**

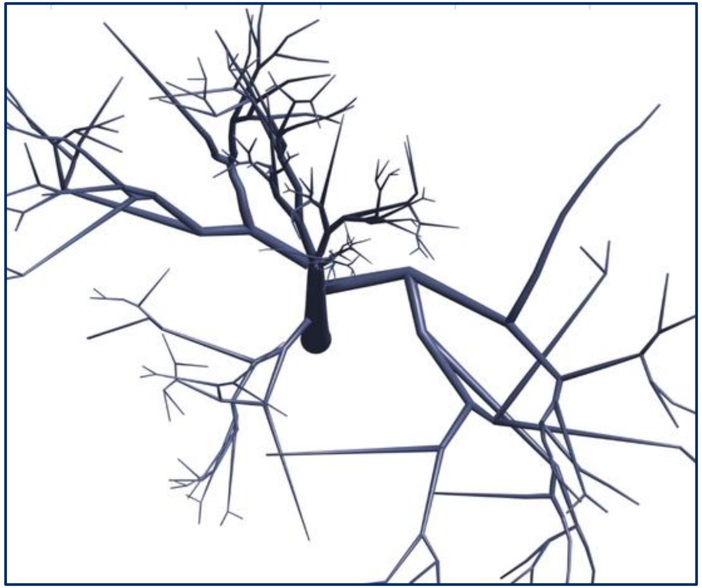

Fig.5: Exemplary QSM (Markku et al., 2015)

# Comparison of Allometries

| Allometry | Application Area | Sample Size | Parameters |
|---|---|---|---|
| **Chave et al., 2014** | Pantropical forests and woodland savannas | 4004 trees from 58 sites (destructive sampling) | DBH + climate stress |
| **Djomo et al., 2016** | African tropical forests | 698 trees from 12 sites (destructive sampling) | DBH + wood density |
| **Terryn et al., 2024** | Pantropical rainforests | 1951 trees from 19 plots (TLS data acquisition) | DBH + MCWD |

Tab.1: Compared DBH-Height-Allometries

| Allometry | Application Area | Sample Size | Parameters |
|---|---|---|---|
| **Chave et al., 2014** | Pantropical forests and woodland savannas | 4004 trees from 58 sites (destructive sampling) | DBH + height + wood density |
| **Djomo et al., 2016** | African tropical forests | 844 trees from 12 sites (destructive sampling) | DBH + height + wood density |
| **Terryn et al., 2024 + Chave et al., 2014** | Pantropical rainforests | 1951 trees from 19 plots (TLS data acquisition) + 4004 trees from 58 sites (destructive sampling) | DBH + MCWD + wood density |

Tab.2: Compared DBH-AGB-Allometries

Application of allometries to Ghanaian trees and comparison with the TLS-derived reference values

- **Accuracy assessment**
  - Root Mean Square Error (RMSE) → magnitude of prediction errors
  - Relative Error (RE) → percentage prediction error
  - Coefficient of Determination ($R^2$) → explained variance and goodness of fit
- **Agreement assessment**
  - Concordance Correlation Coefficient (CCC) → agreement between predicted and observed values
- **Systematic error assessment**
  - Bias → over- or underestimation tendencies
  - Residual Analysis → size-dependent and structural model errors

→ Differentiated evaluation of the different allometries

# Ghana Specific Allometry

- Exploratory data analysis using **scatter plots** (DBH, height, biomass, wood density, environmental variables)
- Comparison of different **model structures** and **predictor combinations**
- Focus on **univariate** and **multivariate power-law models**
- General model form:
  - $Y = aX^b$
  - $Y = aX_1^bX_2^c$
- Log-transformation → linear regression → back-transformation
- Stepwise **model extension** with additional explanatory variables

- Development and comparison of multiple model variants
- Leave-5-out **cross-validation**
- Model comparison:
  - Akaike Information Criterion (AIC)
  - RSE
  - Residual Analysis
- Assessment of model assumptions and systematic errors
- Final model selection:
  - **highest** predictive **performance**
  - **Lowest** necessary **complexity**

# Thank you for your attention